

16. Back End of Compiler

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    char op1, op2, result, op;
```

```
    printf("Enter three address statement.\n");
```

```
    printf("Example: t1 = a + b\n\n");
```

```
    printf("Enter result variable: ");
```

```
    scanf(" %c", &result);
```

```
    printf("Enter first operand: ");
```

```
    scanf(" %c", &op1);
```

```
    printf("Enter operator (+, -, *, /): ");
```

```
    scanf(" %c", &op);
```

```
    printf("Enter second operand: ");
```

```
    scanf(" %c", &op2);
```

```
    printf("\nGenerated Assembly Code:\n");
```

```
    printf("MOV R0, %c\n", op1);
```

```
    switch (op)
```

```
{
```

case '+':

printf("ADD R0, %c\n", op2);

break;

case '-':

printf("SUB R0, %c\n", op2);

break;

case '*':

printf("MUL R0, %c\n", op2);

break;

case '/':

printf("DIV R0, %c\n", op2);

break;

default:

printf("Invalid operator\n");

return 0;

}

printf("MOV %c, R0\n", result);

return 0;

OUTPUT

```
BACK END OF COMPILER
-----
Enter result variable: t
Enter first operand: a
Enter operator (+, -, *, /): +
Enter second operand: b

Three Address Code:
t = a + b

Generated Assembly Code:
MOV R0, a
ADD R0, b
MOV t, R0

-----
Process exited after 30.35 seconds with return value 0
Press any key to continue . . . |
```